SANATH KOUNDINYA

Roll No. 250973 | B.Tech, Electrical Engineering
Indian Institute of Technology Kanpur

sanathk25@iitk.ac.in
+91 97405 00712
GitHub: github.com/Sanath-Koundinya

PROFILE

- Electrical Engineering undergraduate at IIT Kanpur (CPI 9.63, three A* grades, JEE Advanced AIR 1063) with hands-on experience in robot learning research, computational photography and FPGA/HDL design. Strong mathematical background (RMO qualifier, Estimathon 4th place, national and international olympiad ranks) and a record of debugging hard systems problems to root cause.
- Building a public portfolio in SystemVerilog/VHDL (hardware order book, FPGA camera project) and actively completing free certifications to deepen digital-design and low-latency systems skills.

EDUCATION

Year	Degree / Examination	Institute	CPI / %
2025 – 2029	B.Tech, Electrical Engineering (3 A* grades)	Indian Institute of Technology Kanpur	CPI 9.63 / 10
2025	CBSE Class XII (PCM + CS)	Alpine Public School, Bengaluru	JEE Adv. AIR 1063 | JEE Main AIR 3879
2023	CBSE Class X	BGS National Public School, Bengaluru	597/600 (99.5%), State Topper, AIR 3

Also: AP Calculus BC 5/5, AP Computer Science A 5/5, SAT 1530, IELTS 8.5 (C2 Proficiency). A* in Introduction to Electrodynamics.

RESEARCH & WORK EXPERIENCE

Summer Research Intern, WSAIS, IIT Kanpur *Summer 2026*

Robot imitation learning: MimicGen and SkillGen

- Worked on the implementation of SkillMimicGen (CoRL 2024). As no public codebase existed, used its predecessor MimicGen (CoRL 2023) as the base pipeline, running it on robosuite and robomimic.
- Modified the MimicGen codebase and generation scripts to scale automated demonstration generation from a small set of human demos to many synthetic trajectories.
- Peg-in-hole (Square) assembly task: from 10 source demonstrations, ran 200 generation trials and obtained 92 verified successful demos (46% success; mean episode length about 152 steps). Verified demo quality through per-episode success checks, playback videos and episode-length statistics.
- Trained behaviour-cloning policies on source and generated data, and produced before/after rollout comparisons with videos and success counts, reporting only measured results.
- Began the SkillGen implementation in Isaac Sim / Isaac Lab on the lab workstation; prepared and presented the SkillMimicGen paper to the research group.

Undergraduate Project (UGP): Computational Photography *Aug 2026 – Present*

Canonical face representation, albedo and relighting from monocular video; guided by IITK faculty and lab mentors

- **Colour correction and validation.** Learned colour science (albedo, PSNR/SSIM) and corrected ground-truth videos using a SpyderCheckr-48 chart in DaVinci Resolve, working around the tool's 24-patch-only matching. Quantified the result in Python (sRGB → XYZ → Lab, CIE76 ΔE): mean ΔE 15.63 → 11.69, median 16.57 → 8.96, 37 of 48 patches improved, 25.2% mean error reduction.
- **Automated LUT pipeline.** Debugged and fixed my own SpyderCheckr-48 auto-correction pipeline: ffmpeg filtergraph path escaping, fragile audio handling, and a chart-grid detection crash (IndexError from a mismatched correspondence call).
- **FLAME 3D face fitting.** Built an analysis-by-synthesis pipeline (68 landmarks, Adam optimiser, orthographic camera, shape fixed after first frame, temporal smoothness) with a GPU 3D renderer. Diagnosed "shape explosion" as an ill-posed inverse problem (136 constraints vs about 165 parameters); fixed it with regularisation (MAP view) and clamping, reducing max shape and expression magnitudes from 9.6 and 48.6 to 3 or below. Documented the limits of sparse landmarks (no ear, cheek or crown points).
- **State-of-the-art study and runs.** Presented the FLAME paper; set up and ran MICA (identity, 0.13 s/frame), metrical-tracker (about 6.8 s/frame), FlashAvatar (55k iterations, Huber loss down 97%) and RGBAvatar (115 FPS training, 60,353 canonical Gaussians), solving CUDA/gcc build issues and authenticated-asset setup.
- **Batch FLAME extraction at scale.** Benchmarked the tracker for a 30,000+ video dataset; showed visualisation was not the bottleneck (6.5% speed-up) and reverted to preserve accuracy as the advisor required. Root-caused a CUDA out-of-memory crash that blocked a 31,686-video run (an 8.28 GiB face-detector allocation with only 10 to 30 MiB of margin), and fixed it with a 640 px detection step rescaled to full resolution, leaving the tracker code unchanged.
- **Verification discipline.** Validated the fix by bounding-box comparison (about 6 px shift), per-frame FLAME parameter deltas (shape difference exactly 0), TensorBoard loss curves (final photometric loss differed by 1.2%) and end-to-end production runs with zero OOM failures; caught and corrected my own mismatched-identity comparison mid-investigation.
- **Regression report.** Wrote a report on basis-function regression, bias–variance tradeoff and ridge regularisation, with Python simulations of under/overfitting and coefficient shrinkage.

Summer Project: Project FOCI (FPGA Camera), Electronics Club, IIT Kanpur *May – Aug 2026*

Mentee; Verilog/VHDL on the EDGE Spartan-7 (Xilinx XC7S15) board with Vivado

- Part of a from-scratch digital camera project (CMOS sensor, FPGA, serial bus, PCB and optics planning); every module simulated against a testbench before hardware.
- Built and simulated clock dividers (including odd-divide 50/50 duty cycle) and a strobe/enable generator to keep designs in a single clock domain; a UART transmitter FSM at 115200 baud; and an I2C-style master with repeated-START reads checked against a behavioural sensor model (chip-version register 0x1801).
- Board exercises on real hardware: BCD to seven-segment decoder, push-button counter with edge detection, a 16×2 LCD controller in VHDL, an ST7735 TFT driver, and a 3×3 matrix correlation block.
- Studied the working OV7670-to-VGA pipeline: SCCB sensor configuration, pixel capture and subsampling, true dual-port BRAM clock-domain crossing and VGA timing, with MMCM-generated clocks.

PROJECTS

Hardware L2 Order Book in SystemVerilog github.com/Sanath-Koundinya/hw-orderbook

- Synthesizable order book for low-latency trading: 64 price levels per side, 16-bit quantities, add/reduce messages applied in one clock cycle, best bid/ask from per-side occupancy masks.
- Verified with cocotb: directed tests plus 5000 random messages compared with a Python reference model after every message; Verilator lint clean.
- Yosys synth_xilinx: flip-flops reduced from 2176 to 128 by resetting only the masks so quantities map to distributed RAM. Simulation and synthesis only (no hardware run yet); the 64-wide priority encoder is the likely critical path. Planned: registered outputs, ITCH-style parser, per-order tracking.

EyeSeeYou: Smart Surveillance System *Raspberry Pi, YOLO, OpenCV, OCR, MongoDB*

- Real-time vehicle detection and licence-plate recognition on a Raspberry Pi with motion-triggered recording; YOLO plus OCR post-processing picks the best frame per event.
- Multithreaded architecture running capture, inference, AES-256 encryption and MongoDB GridFS storage concurrently into an encrypted, searchable backend.

Giving Hub App *MIT Summer Appathon 2023*

- Mobile platform connecting volunteers with local service opportunities, built in MIT App Inventor for the competition.

SCHOLASTIC & MATHEMATICAL ACHIEVEMENTS

- **Estimathon:** 4th place with a two-member team; first among Y25 teams in our round.
- **Regional Mathematical Olympiad (RMO):** qualified twice. IOQM Certificate of Merit (Centre Top 10%); NMTC Stage 2 qualifier (2024); Aryabhatta Ganit Challenge.
- Unified International Mathematics Olympiad National Rank 30 (Top 100); SOF International Mathematics Olympiad International Rank 33 (Level 1) and 60 (Level 2); SOF National Science Olympiad International Rank 15 (Medal of Distinction); NSEP Certificate of Merit (Centre Top 10%).
- JEE Advanced AIR 1063 (top 0.1% of about 1.8 million candidates); JEE Main AIR 3879; CBSE Class X Karnataka State Topper, AIR 3 nationally.
- 1st Place (joint 3rd), Electrovista Hackathon (Electronics Club); Winner, AstroVenture (Astronomy Club); Bronze Medal, Inferno Carrom Tournament.

TECHNICAL SKILLS

- **Hardware / HDL:** SystemVerilog, Verilog, VHDL, cocotb, Verilator, Icarus Verilog, Yosys, Xilinx Vivado, Arduino, Raspberry Pi.
- **Programming and tools:** Python, C, PyTorch, OpenCV, ffmpeg, Git, Linux, MongoDB, TensorBoard; robosuite, robomimic, Isaac Sim / Isaac Lab; DaVinci Resolve.
- **Mathematics:** Linear Algebra, Differential Equations, Discrete Mathematics, probability and estimation, regression.
- **Coursework:** Fundamentals of Computing I & II, Quantum Physics, Introduction to Electrodynamics (A*), Signals, Systems and Networks (EE200).
- **Learning Undertaking:** Currently completing free online certifications in digital design, FPGA prototyping and low-latency systems, and building further SystemVerilog/VHDL projects on GitHub. Committed to learning whatever tools or domains a role requires.

POSITIONS OF RESPONSIBILITY

Secretary, Debate Society (DebSoc), Student Gymkhana *Jan 2025 – Present*

- Run debate events, inter-collegiate competitions and training sessions; represented IITK at NSMPD, Dialectica, VIT Pre-UADC and Charcha.

Secretary, Book Club, Media & Culture Wing *Aug 2025 – Present*

- Organise reading initiatives, literary discussions and author talks for the campus.

Core Group Member, Vox Populi *Jan 2026 – Present*

- Editorial and creative direction for a prominent student publication.

Founding Member, Animal Allies, Community Welfare Cell *Aug 2025 – Present*

- Part of the founding team under the Student Gymkhana; led proposal drafting for formal recognition; wrote campaign captions and slogans and took part in Know Your Campus outreach.

ARTS & CULTURE: CARNATIC MUSIC

- **Diploma in Carnatic Music, Chandigarh University.**
- Grade A Double Star, KalaSourabha 2023 International Music & Dance Competition (Carnatic Vocals); Finalist, Jhankriti 2023, World Forum for Art and Culture.
- High Distinction (96.5%), Karnataka State Education Examination Board, Carnatic Classical Vocal Junior Exam.

VOLUNTEERING & INTERESTS

- Student Volunteer, Ramakrishna Mission (May 2022 – Present): led a school-supplies drive for village children (Shivanahalli) and translated Swami Vivekananda's Quiz from English to Kannada.
- COVID-19 Community Tutoring (2020 – 2021): taught Mathematics and Science to children of domestic and security staff.
- Languages: English (IELTS 8.5, C2), Hindi, Kannada, Sanskrit. Interests: competitive debating, reading, badminton, gardening and community plantation drives.